

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination 2025-26 (NEP)

DEPARTMENT OF GEOLOGY**Paper Code: ESMJ43 / ESMN41 — Environmental Geology and Hydrogeology**

Time: 2:30 Hours

Maximum Marks: 70

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Answer five questions in all, selecting at least two questions each from Section–A and Section–B. Section–C is compulsory. All questions carry equal marks (14 marks each).***SECTION – A****Question 1** [14 Marks]

Explain how global climate drivers, orbital parameters, and surface albedo feedback loops modulate global temperatures.

Question 2 [14 Marks]

What is Darcy's Law? Derive its equation and comment on the validity of Darcy's Law with respect to Reynolds number.

Question 3 [14 Marks]

What is the hydrological cycle? Explain briefly the various components of the Hydrological Cycle with a neat sketch.

SECTION – B**Question 4** [14 Marks]

Briefly discuss the subsurface methods of groundwater exploration.

Question 5 [14 Marks]

Describe the basic terms related to aquifer properties. Calculate the groundwater storage for an aquifer having an areal extent of 10 km^2 and a saturated thickness of 15 m. Consider the specific yield of the aquifer to be 20%.

Question 6 [14 Marks]

Attempt any **two** of the following ($2 \times 7 = 14$ Marks):

- (i) Hydro-chemical properties used to characterize groundwater quality
- (ii) Causes of groundwater depletion
- (iii) Geological mechanism of earthquake generation and seismic hazards

- (iv) Landslides and their causes

SECTION – C (Compulsory)

Question 7

[14 Marks]

Answer the following in 2 to 3 sentences each ($7 \times 2 = 14$ Marks):

- (i) Environmental legislation
- (ii) Role of Stratosphere
- (iii) Population growth
- (iv) Aquifer
- (v) Water table and Water level
- (vi) Intrinsic Permeability
- (vii) Effluent and Influent conditions